2.

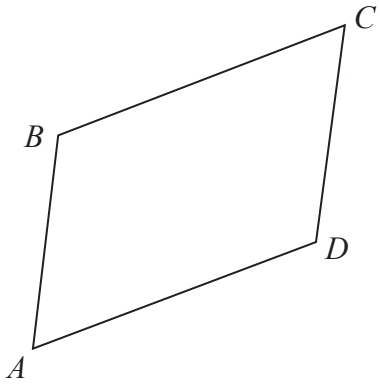


Figure 1

Figure 1 shows a sketch of parallelogram $ABCD$.

Given that $\overrightarrow{AB} = 6\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$ and $\overrightarrow{BC} = 2\mathbf{i} + 5\mathbf{j} + 8\mathbf{k}$

(a) find the size of angle ABC, giving your answer in degrees, to 2 decimal places.

(3)

(b) Find the area of parallelogram $ABCD$, giving your answer to one decimal place.

(2)